



NA.YADRA Project Capability Summary

Pacific Maritime Intelligence Platform

This document summarises the current capability of the NA.YADRA maritime intelligence dashboard from its early build stage to its present operational prototype. The platform is designed to support Pacific maritime domain awareness by combining AIS vessel tracking, behavioural detection, risk scoring, satellite-style map overlays, cyber intelligence indicators, and analyst reporting.

1. Current Project Capability

Capability	What it does now
Live AIS vessel tracking	Displays vessels around Fiji and the wider Pacific using AISStream data.
Risk scoring	Scores vessels as LOW, MEDIUM, or HIGH based on behaviour patterns.
Behavioural detection	Detects night movement, offshore low speed/stopped vessels, possible night stop locations, repeat activity.
Port proximity logic	Compares vessel location against known port zones such as Lautoka/Vuda, Denarau/Nadi, and Suva Harbour.
Dashboard map	Shows vessels on an interactive map with satellite and street-map style overlays.
Intel summary panel	Summarises total vessels, high-risk vessels, medium-risk vessels, low-risk vessels, and stale contacts.
Cyber feed health panel	Tracks cyber feed status such as OpenPhish integration and last check time.
Night movement report	Generates PDF intelligence reports for vessels detected moving during Fiji night hours.
Possible night stop report	Generates PDF reports for vessels stopped or slow offshore during Fiji night hours.
Analyst notes	Allows suspicious vessel behaviour to be written up as intelligence leads.

2. Dashboard Panels Explained

Panel	Purpose
Map Panel	Main operational view showing vessel locations, port zones, alert markers, and movement patterns.
Vessel Cards	Lists vessels with risk level, score, offender status, speed, and behavioural flags.
Intel Summary	Gives a quick operational snapshot of vessel risk across the monitored area.
Cyber Feed Health	Shows whether cyber threat feeds are active and when they were last checked.
Alert / Behaviour Popups	Explains why a vessel was flagged, including MMSI, score, nearest port, distance, coordinates, and behaviour.
PDF Report Output	Creates exportable reports suitable for briefing, analyst review, or sharing with trusted maritime contacts.

3. Operational Value Today

NA.YADRA can already act as a prototype maritime intelligence support tool. It can highlight vessels behaving unusually near Fiji waters, especially at night, near port approaches, or outside normal port zones. It does not replace official systems, but it can provide early warning leads, visual intelligence summaries, and structured reports for analyst review.

4. Future Capability With Stronger Satellite Fusion

Future Upgrade	What it could add
Higher-resolution satellite imagery	Better visual confirmation of vessels, small craft, anchorages, and remote coastal activity.
More frequent satellite refresh	Improved ability to compare AIS position against real-world imagery closer to real time.
AIS dark activity detection	Detect vessels that disappear from AIS and later reappear in suspicious areas.
Satellite-AIS mismatch detection	Flag cases where AIS says one thing but imagery suggests another.
Small craft / mothership detection	Support detection of possible rendezvous patterns between larger vessels and smaller craft.
Historical pattern analysis	Build stronger behaviour profiles across days, weeks, and months.
Automated intelligence briefings	Generate daily reports for agencies such as customs, navy, ports, and border security.
Regional Pacific expansion	Extend coverage to Samoa, Tonga, Vanuatu, Tuvalu, Kiribati, PNG, Micronesia, and wider Pacific routes.

5. Why Innovate UK Funding Would Matter

Innovate UK support could help move NA.YADRA from a working prototype into a stronger pilot platform. Funding could support improved satellite data, cloud deployment, secure user accounts, better data storage, professional reporting, alert automation, and regional testing with trusted partners.

6. Important Note

All detections are intelligence leads for review only. A flagged vessel does not confirm criminal activity. The system is designed to support maritime awareness, prioritisation, and further investigation by authorised analysts.